Beyond Motion Generation

From Motion Execution to Visuomotor Learning

Group 2

What if we could learn from the motions we generate, rather than just execute them?

The University of Hong Kong
DASC7606C | Track 4: Humanoid-robot Motion Generation

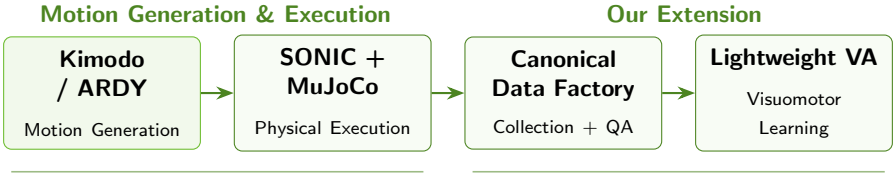
October 2026



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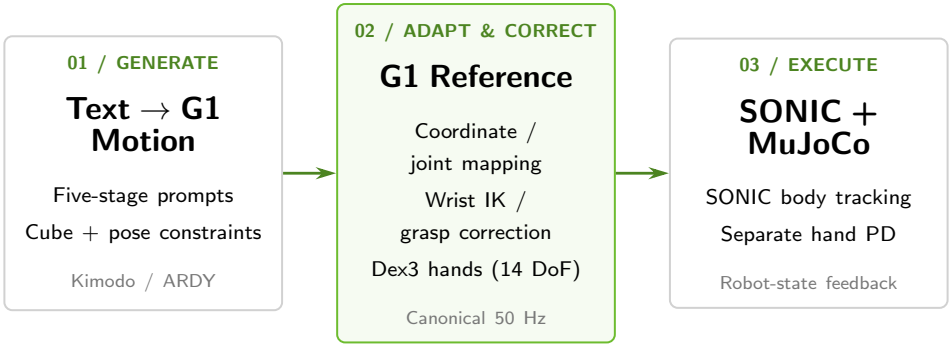
What if we could learn from the motions we generate, rather than just execute them?



From Text to Physical Grasping

Two offline motion generators

One shared SONIC execution pipeline



DEMONSTRATIONS

Two separate 16-second videos

▷ **Kimodo Demo**

▷ **ARDY Demo**

ARDY and Kimodo Pipeline Comparison

PIPELINE · SHARED INPUT AND EXECUTION, DIFFERENT GENERATION … CTION

▷ Watch Demo

Task and constraints

Same text task

Same wrist and pose targets

Same center setting

Motion generation

ARDY
Segmented diffusion motion

KIMODO
Segmented constraint generation

Adapter and correction

ARDY
Frozen ARDY adapter
Center tracking calibration

KIMODO
Canonical adapter
Contact and approach refinement

Controller and replay

810 frames at 50 Hz

SONIC body control

Hand PD and MuJoCo

Success validation

Lift at least 0.10 m

Hold at least 2.0 s

Complete synchronized evidence

Comparison object

Frozen end-to-end generation and adaptation pipelines, not raw generators alone

Both pipelines publish the same body and hand reference interface to the shared controller.

▷ Watch Demo

Formal40 Result

RESULT · FIXED CENTER SETTING, SEEDS 1000–1019, ONE CANDIDATE PER SEED

Verified success

19/20 each pipeline

ARDY 95% **Kimodo 95%**

Median end-to-end time

Kimodo **162.7 s**

ARDY **182.7 s**

Failure cases

Kimodo seed 1006	Physical task failure: no valid grasp, lift, or hold
ARDY seed 1007	Physical lift succeeded, but replay evidence integrity failed

Same verified reliability in this fixed setting; different failure modes and runtime profiles

Timing is descriptive because the runs used different MTT S4000 device IDs.

Bonus1:

Exploring VLM-Guided Motion Planning for Robotic Manipulation

Attempt 1: Real-time Visual Prompting

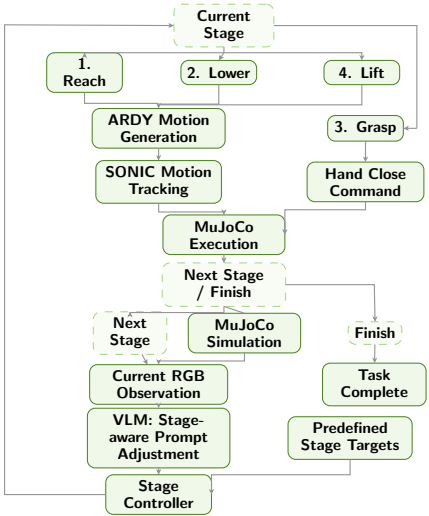
Attempt 2: Stage-wise VLM Agent

Attempt 3: Pixel-level Visual Grounding

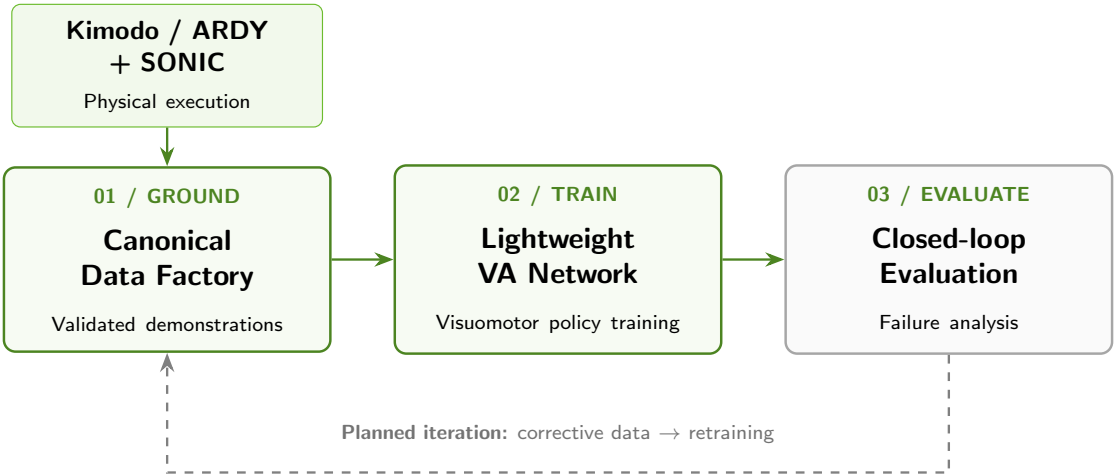
Final Approach: Hybrid VLM-Guided Control

final approach

The task remains divided into four predefined stages. Each stage uses a predefined spatial target to maintain motion consistency, while the VLM processes the current simulation image and adjusts the stage-specific text prompt.

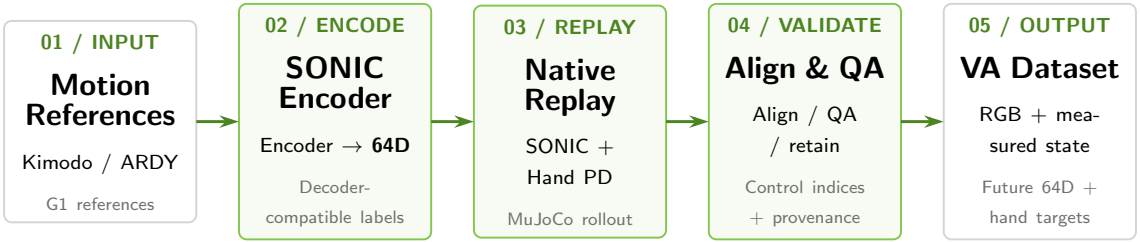


Towards an Iterative Learning System



Canonical Data Factory — System Overview

Generated G1 motion is **not yet VA training data**. We convert it into **SONIC-native targets** paired with physical observations.



CONTROL REPRESENTATION

64D body tokens + 14D hand references

SONIC-native supervision for VA training

PHYSICAL GROUNDING

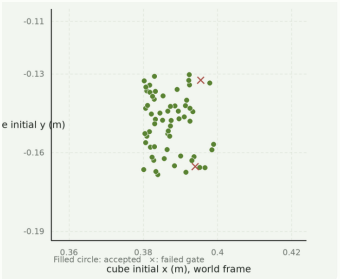
RGB + real simulated robot state

Grasp QA is separate from token and temporal QA

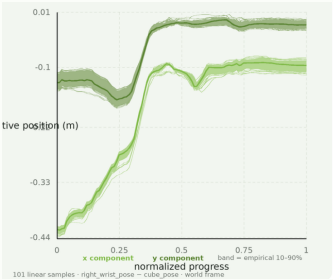
Inside the Data Factory — Production & Dataset Inspection

Batch Replay → SONIC Replay → Data Capture → Align + QA → VA Episodes

01 / Cube placement (60 accepted)

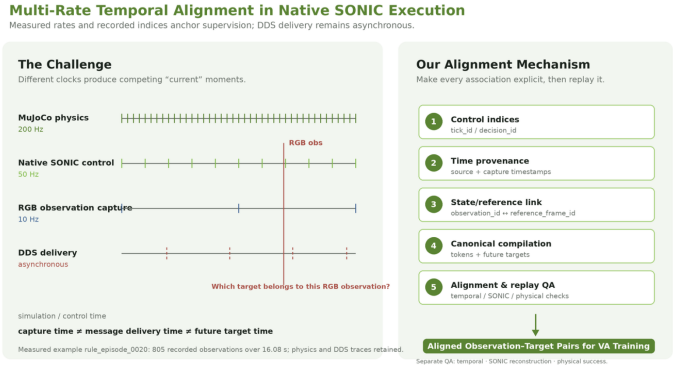


02 / Expert wrist trajectories



Inspection: 60 accepted right-hand episodes span a local workspace. Narrow expert wrist bands do not establish closed-loop robustness.

Engineering Challenge — Multi-Rate DDS Alignment



SECONDARY / SONIC BACKEND PORTING

TensorRT $\rightarrow$ MUSA (MTT S4000); $\sim$ 11 s Policy Session setup.

From Accurate Predictions to Reliable Grasping

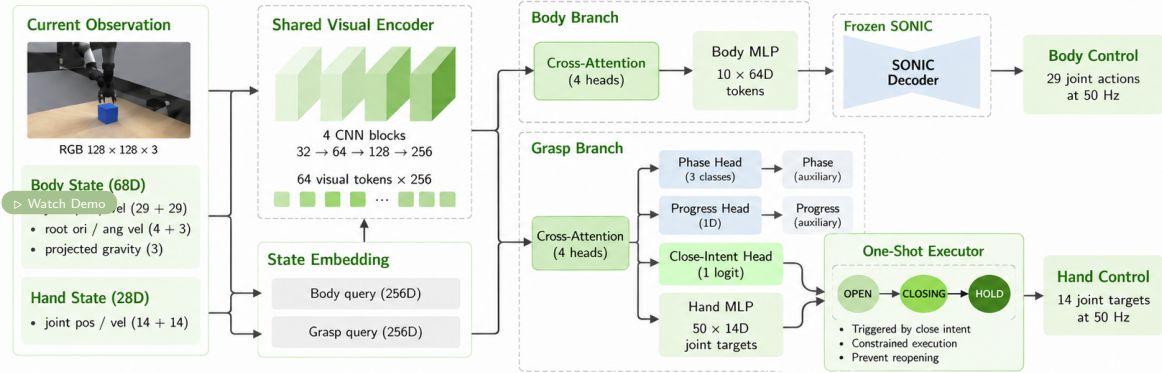
Version	What We Changed	What We Learned
V1	Explicit joint-trajectory prediction	SONIC needs velocity and root pose; direct joint targets need approximations.
V2	Native 64D tokens with state history	High offline accuracy; learner-state feedback favored hand continuation.
V3	Remove state history; shorten horizon	Grasp transitions still failed; faster inference faced real-time constraints.
V4.0	Spatial cross-attention + grasp / phase heads	Auxiliary phase predictions did not prevent repeated closing and reopening.

Motion Representation $\longrightarrow$ **Distribution Shift** $\longrightarrow$ **Grasp Transition**

Prediction accuracy alone does not ensure temporally consistent grasp execution.

▷ **Failure clip**

Final VA Architecture — BCZ-Hand



▷ Watch Demo

Training Configuration

Model size	3.52M trainable parameters
Training data	9,000 samples
Training epochs	20 epochs
Optimizer	AdamW (lr 3e-4, batch 32)
Training loss	Body + Hand + Phase + Progress + Close-Intent + Future Progress

Training Objective

$$\mathcal{L} = \mathcal{L}_{body} + 3\mathcal{L}_{hand} + 0.1\mathcal{L}_{phase} + 0.1\mathcal{L}_{progress} + 0.2\mathcal{L}_{close} + 0.5\mathcal{L}_{advance}$$

- Body: 10 × 64D SONIC tokens (0.20 s)
- Hand: 50 × 14D joint targets (1.00 s, +40 ms lead)
- Close intent: binary classification
- Phase / progress: auxiliary supervision
- Future progress: encourage sufficient closure

Closed-Loop Result (Simulation)

Maximum lift height	Valid hold duration
12.50 cm	**3.24 s**

Beyond Motion Generation

From executing generated motion to learning from it.

01	EXECUTE	**Kimodo / ARDY** G1 grasp via SONIC; 19/20 verified each (fixed center).
02	COMPILE	**Canonical Data Factory** Automated replay, indexed data, physical QA and dataset export.
03	LEARN	**Lightweight VA** Rule-trained; one grasp interval: 12.5 cm lift / 3.24 s hold.

NEXT **Robustness, repeatability and a closed data flywheel**
New scenes • Repeatable grasps • Evaluation → correction → retraining (future)

VA evidence is one valid physical grasp interval; repeatability and terminal-height success are not yet established.

▷ Watch Demo

Acknowledgments

Thank You!

Questions & Discussion

Our thanks to the course instructors and the project team.

PROJECT REPOSITORY
github.com/hkuDL/dl-group2

Beyond Motion Generation | Group 2 | HKU Track 4